

AI Veterinary Assistance: Enhancing Clinical Decision-making in Animal Healthcare

Guide: Ms. Sheelu Susan Mathews

Project Domain: Artificial Intelligence

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Introduction

- Increasing companion animal ownership and chronic disease prevalence have amplified demand for veterinary services.
- Veterinarians face heavier workloads and shorter consultation times, reducing diagnostic accuracy and owner satisfaction.
- **AVA (AI Veterinary Assistance)**: an LLM-driven framework that extracts symptoms, predicts candidate diseases using a veterinarian-certified disease–symptom database, and recommends follow-up questions to support clinical decision-making.

Problem Statement

- Veterinary consultations are often unstructured and overloaded, making symptom extraction and early differential diagnosis time-consuming.
- Existing automatic tools focus on single diseases/procedures; there is a lack of explainable, database-backed consultation assistants for veterinarians.
- Need: a lightweight, explainable system to assist veterinarians in early-stage consultations by extracting symptoms and suggesting candidate diseases/questions.

Project Objectives

- 1 Build AVA to automatically extract symptoms from consultation notes using LLM-based in-context learning.
- 2 Use a veterinarian-certified disease–symptom database to match symptoms and predict Top-K candidate diseases.
- 3 Recommend focused follow-up consultation questions (most-frequent / most-decisive strategies).
- 4 Provide transparent intermediate outputs (extracted symptoms, candidate narrowing) for explainability.
- 5 Design a maintainable database-driven system suitable for general veterinary clinics.

Project Scheduling

- **Month 1:** Detailed literature survey, dataset & database consolidation.
- **Month 2:** Prototype symptom-extraction prompts + database ingestion.
- **Month 3:** Disease-matching & Top-K prediction module; question recommendation logic.
- **Month 4:** Integration, UI for vet review, and explainability outputs.
- **Month 5:** Testing on generated + real-world records, evaluation, documentation.

SDG 3 — Good Health and Well-being

- Improves diagnostic support and clinical outcomes for animals; supports broader animal welfare.

SDG 9 — Industry, Innovation and Infrastructure

- Introduces low-cost, scalable AI tools for clinical decision support in veterinary settings.

SDG 10 — Reduced Inequalities

- Enables improved veterinary support in resource-limited areas through augmentation tools.

Paper 1 — Large Language Models are Few-Shot Clinical IE (EMNLP 2022)

Title: Large Language Models are Few-Shot Clinical Information Extractors

Authors: Agrawal et al., EMNLP 2022

Objectives:

- Demonstrate general-purpose LLMs can perform clinical IE with zero/few-shot examples.
- Introduce guided prompt design and datasets for benchmarking clinical extraction.
- Provide a Prompt+Resolver workflow for structured outputs.
- Address the challenge of extracting structured information from unstructured clinical text.
- Reduce the need for large labeled datasets in clinical NLP tasks.

Paper 1 — System Architecture

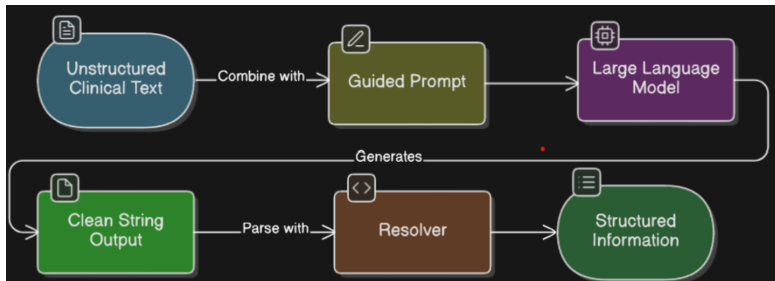


Figure: Prompt + Resolver workflow: zero-shot vs guided one-shot example.

Paper 1 — Design Methodology, Implementation Strategy

Key Components:

- **Guided Prompt Design:** One-shot example enforces predictable output format
- **Resolver Function:** Lightweight parser (under 10 lines of code) maps LLM text to structured data
- **New Benchmarks:** Three annotated datasets from CASI for coreference, medication status, attribute extraction

Implementation Details:

- Used InstructGPT as the base LLM accessed via API
- Prompt template includes task description, example, and input
- Resolver handles key-value pair extraction from LLM output
- Model distillation pipeline for local, privacy-safe deployment

Paper 1 — Results, Comparison & Future Scope

Key Results:

- **Biomedical Evidence Extraction:** 85% accuracy vs 35% for PubMedBERT-CRF
- **Clinical Sense Disambiguation:** 86% accuracy, outperforming 71% SOTA
- Few-shot LLM matched or outperformed supervised baselines on 5 clinical NLP tasks
- Significant reduction in labeled data requirements

Future Scope:

- Explore local LLMs (e.g., OPT) for privacy-preserving applications
- Incorporate uncertainty estimation for reliable deployment
- Extend to multilingual clinical text processing
- Develop domain-specific resolvers for different clinical specialties

Paper 2 — RepVGG for X-Ray Classification (arXiv 2022)

Title: Artificial Intelligence for Automatic Detection and Classification of Disease on X-Ray Images

Author: Liora Mayats-Alpay (arXiv, 2022)

Objectives:

- Present RepVGG-based architecture for fast and accurate X-ray classification.
- Classify chest X-rays into COVID-19, Pneumonia, Normal.
- Leverage structural re-parameterization for training-time complexity and inference-time simplicity.
- Provide explainable results using Grad-CAM visualization.
- Address the need for rapid diagnosis in critical respiratory diseases.

Paper 2 — System Architecture

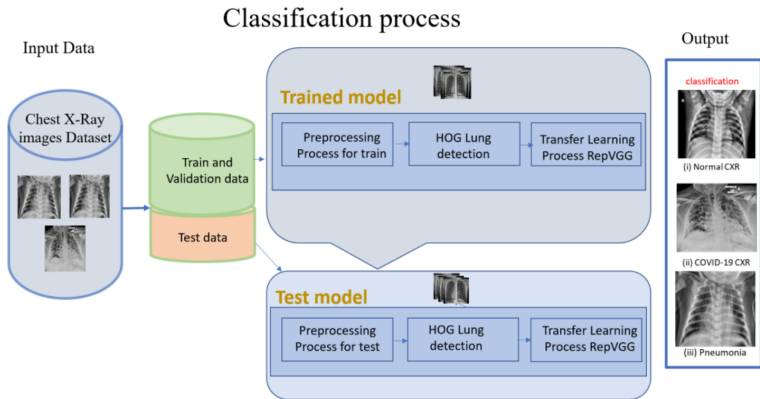


Figure: RepVGG training-time multi-branch → inference-time plain architecture.

Paper 2 — Design Methodology & Implementation

Dataset Details:

- Public Kaggle dataset with 2,856 chest X-ray images
- Distribution: COVID-19 (1,000), Pneumonia (856), Normal (1,000)
- Split: 80% training/validation, 20% testing

Preprocessing Pipeline:

- Image resizing to standard dimensions
- Rotation augmentation for robustness
- HOG-based lung detection for ROI extraction
- Normalization of pixel values

Training Strategy:

- Transfer learning from pretrained RepVGG
- Structural re-parameterization during training
- Single-path architecture during inference for speed

Paper 2 — Results, Comparison & Future Scope

Performance Metrics:

- **Overall Accuracy:** 95.79%
- **COVID-19 Classification:** 94.59% (vs ResNet50: 84.13%)
- Outperformed VGG-16, ResNet50, InceptionV3, DenseNet
- Grad-CAM heatmaps showed focus on infected lung areas

Key Advantages:

- High accuracy with fast inference speed
- Memory-efficient during deployment
- Explainable through attention visualization

Future Scope:

- Adapt to other medical imaging modalities (MRI, CT)
- Mobile deployment for field diagnostics
- Multi-disease classification framework
- Integration with clinical decision support systems

Paper 3 — 1D-CNN Lung Sound Classification (Sensors 2024)

Title: Towards Clinical Decision Support via Lung Sound Classification using 1D-CNN

Authors: Ali S.W., Rashid M.M., Yousuf M.U., Shams S., et al. (Sensors, 2024)

Objectives:

- Build 1D-CNN for lung sound classification (normal/crackles/wheezes/combined).
- Address class imbalance using SMOTE technique.
- Provide computationally efficient model for real-time applications.
- Achieve high accuracy in distinguishing respiratory conditions.
- Enable deployment on resource-constrained medical devices.

Paper 3 — System Architecture

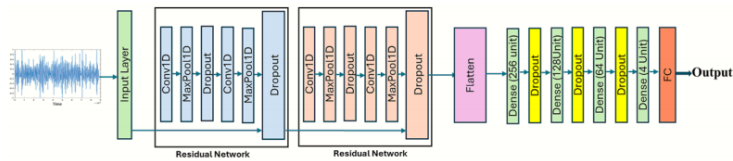


Figure 6. One-Dimensional Convolutional Neural Network (1D-CNN) architecture.

Figure: 1D-CNN architecture with 4 Convolutional Layers and Dense Output Layer.

Paper 3 — Design Methodology & Implementation

Dataset: ICBHI 2017 Breathing Sound Database

- 920 audio samples, 6,898 respiratory cycles
- Distribution: Normal (3,642), Crackles (1,864), Wheezes (886), Combined (506)
- 128 participants, 5.5 hours of recordings

Preprocessing Pipeline:

- Resampling to 4 kHz uniform rate
- Segmentation into 20-25ms samples
- Spectrogram validation for quality control
- Removal of invalid segments without crackles/wheezes

Class Imbalance Handling:

- SMOTE applied to training data
- Synthetic sample generation to balance classes
- Improved performance on minority classes

Paper 3 — Results, Comparison & Future Scope

Performance Results:

- **Overall Accuracy:** 95%
- **F1-Scores:** Normal (0.97), Crackles (0.95), Wheezes (0.94), Combined (0.89)
- **Cross-validation:** 5-fold (97.1%), 10-fold (97.3%)

Comparison with SOTA:

- Outperformed CNN-MoE (93%) and Co-tuning+SN (94%)
- Computationally efficient compared to 2D-CNN approaches
- Strong performance on minority classes due to SMOTE

Future Scope:

- Ablation studies for model components
- Integration with real-time clinical systems
- Deployment on portable medical devices
- Extension to other respiratory conditions
- Multi-language adaptation for global use

Paper 4 — AI for Enhancing Veterinary Healthcare (Experiment)

Title: Artificial Intelligence for Enhancing Veterinary Healthcare: An Experiment

Objectives:

- Apply DenseNet-121 to detect Lumpy Skin Disease (LSD) in cattle.
- Evaluate cross-breed generalization and field deployment potential.
- Support SDG 3 (Health), SDG 9 (Innovation), SDG 15 (Life on Land).
- Develop scalable deep learning architecture for veterinary healthcare.
- Enhance early detection of animal diseases through AI-based diagnostics.

Paper 4 — System Architecture

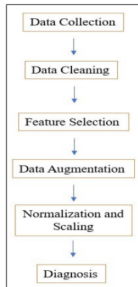


Figure: DenseNet-based classification pipeline for LSD detection.

Paper 4 — Design Methodology & Implementation

Dataset Composition:

- Multi-breed cattle: Vechur (450), Swiss Holstein (803), Jersey (500), Ponwar (450)
- Total: 2,203 images across four breeds
- Balanced representation for cross-breed validation

Preprocessing Pipeline:

- Image resizing to 224×224 pixels
- Normalization of pixel values
- Data augmentation (rotation, flip, brightness)
- Binary classification: Lumpy vs Normal

Model Configuration:

- DenseNet-121 with pretrained ImageNet weights
- Adam optimizer with categorical cross-entropy loss
- 80-20 train-validation split, 25 epochs

Paper 4 — Results, Comparison & Future Scope

Performance Metrics:

- **Training Accuracy:** 99.76%
- **Validation Accuracy:** 92.17%
- **Overall Accuracy:** 93.2%
- **AUC Score:** 0.96

Class-wise Performance:

- **Lumpy Class:** Precision: 96.43%, Recall: 81.82%, F1: 88.52%
- **Normal Class:** Precision: 92.00%, Recall: 98.57%, F1: 95.17%

Future Scope:

- Extend to other animal diseases and species
- Incorporate real-time monitoring systems
- Develop mobile applications for field veterinarians
- Address dataset limitations and improve model interpretability
- Multi-disease models for comprehensive veterinary care

Paper 5 — Predicting Disease Onset from EHR (Frontiers AI 2024)

Title: Predicting disease onset from electronic health records for population health management

Authors: Grout R., Gupta R., Bryant R., Elmahgoub M.A., et al. (Frontiers in AI, 2024)

Objectives:

- Scalable deep learning on longitudinal EHRs for disease onset prediction.
- Integrate explainability (SHAP) and evaluate transfer across populations.
- Support SDG 3 (Health), SDG 9 (Innovation), SDG 10 (Reduced Inequalities).
- Predict chronic diseases: Diabetes, COPD, Hypertension, Myocardial Infarction.
- Consider social determinants for fair predictive healthcare.

Paper 5 — System Architecture



Figure: EHR → embedding → Bi-GRU → pooling → classifier + SHAP architecture.

Paper 5 — Design Methodology & Implementation

Dataset: Accenture's AHA Platform

- Approximately 50 million patients across 39 U.S. healthcare networks
- Longitudinal EHRs with diagnoses, medications, lab results, demographics
- Pretrained embeddings on 30 million records, fine-tuned for prediction

Data Processing:

- Standardized coding: SNOMED-CT, LOINC, NDC
- Normalization and binning of lab test values
- Word2Vec (Skip-Gram) for medical code embedding
- Bi-GRU network with Global Max Pooling

Explainability:

- SHAP values for feature importance
- Identifies most influential factors for predictions
- Transparent decision-making for clinical trust

Paper 5 — Results, Comparison & Future Scope

ROC-AUC Performance:

- **Diabetes:** 0.92
- **COPD:** 0.94
- **Hypertension:** 0.92
- **Myocardial Infarction:** 0.94

Key Findings:

- Adding social determinants and lab values improved prediction by 3.6%
- Comparable to BEHRT and Hi-BEHRT architectures
- Strong transfer learning benefits in limited-data scenarios
- Fairness considerations across demographic subgroups

Future Scope:

- Incorporate multi-country EHR datasets for global applicability
- Deploy predictive tools into real-world clinical systems
- Address fairness and bias in AI-driven healthcare
- Include more social determinants of health
- Real-time risk scoring for preventive care

Comparison Part 1: Text & Record-Based Frameworks

AVA (Veterinary)

Task: Real-time decision support for veterinarians.

Data: Consultation records (text).

Strength: Explainable and database-grounded.

GPT-3 (Resolved)

Task: Extract structured information from clinical notes.

Data: Clinical notes (text).

Strength: Flexible few-shot learning for text understanding.

GRU + Word2Vec

Task: Long-term (3-year) disease risk prediction.

Data: Structured EHR (text + coded fields).

Strength: Scalable for population-level analysis.

Comparison Part 2: Image & Signal-Based Frameworks

RepVGG

Task: Classify diseases from X-ray images.

Data: Chest X-ray images.

Strength: High-accuracy image analysis.

DenseNet-121 (Veterinary)

Task: Detect Lumpy Skin Disease in cattle.

Data: Cattle skin lesion images.

Strength: Precise visual diagnosis in veterinary use-cases.

1D-CNN

Task: Classify respiratory conditions from lung sounds.

Data: Lung sound recordings (audio).

Strength: Efficient feature extraction from signals.

AI Frameworks at a Glance: A Readable Comparison

HeaderColor		2RowColorLightwhite		
RowColorDark	Veterinary	Real-time decision support	Consultation records (text)	Explainable and database-grounded
AVA				
GPT-3 (Resolved)	Human	Information extraction	Clinical notes (text)	Flexible few-shot learning for text understanding
GRU + Word2Vec	Human	Long-term risk prediction	Structured EHR (text + coded fields)	High scalability for population-level analysis
RepVGG	Human	Disease classification	Chest X-ray images	High-accuracy image analysis
RowColorDark	Veterinary	Disease classification	Cattle skin lesion images	Precise visual diagnosis for veterinary use
DenseNet-121				
1D-CNN	Human	Disease classification	Lung sound recordings	Efficient signal processing and

Individual Contributions

- **Alen Denny:** Database design, symptom-schema normalization, dataset curation.
- **Basil Paul:** Audio preprocessing experiments, 1D-CNN baseline and tests.
- **Christin V S:** Literature review, slides compilation, comparative analysis.
- **Christo Tomy Joseph:** Backend integration, evaluation scripts, prototype UI.

Prototypes Developed - Individual

Alen Denny:

- Disease-symptom database schema
- Symptom normalization pipeline

Basil Paul:

- Audio preprocessing module
- 1D-CNN baseline implementation

Christin V S:

- Literature survey compilation
- Comparative analysis framework

Christo Tomy Joseph:

- Backend API integration
- Prototype UI for veterinarian review

Proposed System

Modules:

- Data acquisition (consultation notes, images, audio).
- Symptom extraction (LLM in-context prompt + resolver).
- Disease-matching (database lookup, frequency + coverage ratio ranking).
- Question recommender (most-frequent / most-decisive strategies).
- Vet review UI (explainability: extracted symptoms, matched symptoms per disease).

Proposed System Architecture (AVA)

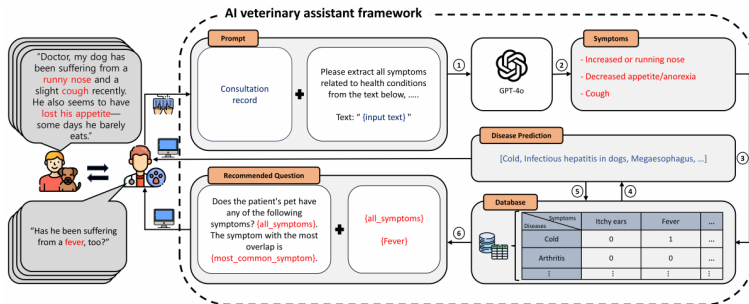
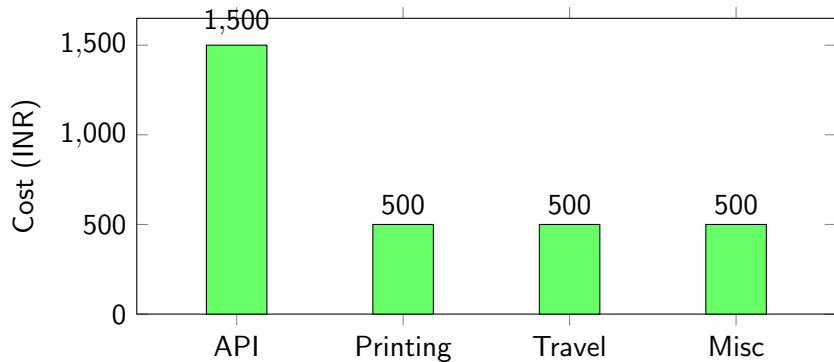


Figure: AVA overview: Consultation input → Symptom extraction (LLM) → DB matching → Top-K ranking → Question recommender → Vet UI.

Project Budget - Minimal Estimation

Item	Cost (INR)
Cloud API credits (free tier + minimal)	1,500
Free domain (subdomain)	0
Free hosting (GitHub Pages)	0
Open-source software	0
Printing (project report only)	500
Local travel (public transport)	500
Miscellaneous (stationery)	500
Total	3,000

Budget Distribution



Cost-Saving Strategies

- **Free Resources:**

- GitHub Student Developer Pack (free credits)
- OpenAI API credits for academic research
- Google Colab for model training
- Free datasets (Kaggle, UCI, etc.)

- **University Resources:**

- Lab computers for development
- University internet and power
- Library access to research papers
- Department printing facilities

- **Open-Source Tools:**

- VS Code, Python, R (free)
- SQLite for database (free)
- Streamlit for UI (free)
- Git for version control (free)

Identified Funding Sources

- **Personal Contribution:** 1,500 (team contribution)
- **Department Fund:** 1,000 (minor project support)
- **Free Credits:** Equivalent to 500 (API credits)
- **Total Available:** 3,000

Conclusion

- AVA integrates LLM-based symptom extraction with a veterinarian-certified disease–symptom database to support real-time consultations.
- The database-driven architecture enables updates without retraining, improving maintainability and transparency.
- Future steps: improve negation handling, localization (language variants), vet-in-the-loop UI, and field testing with clinics.

References



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